

Health & Safety

Standard practices to secure access and work at height



The standard practices for securing access to height and work at height presented in this document serve as a practical guide to define the means to be planned and implemented to ensure effective protection against the risk of falling.

They do not supersede the applicability of the reference document **Standard Fall from height Prevention (BYTP-H&S-REF-2221)**.

The elements of this document can also serve as a support for the briefing of the various personnel involved in a work at height operation.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

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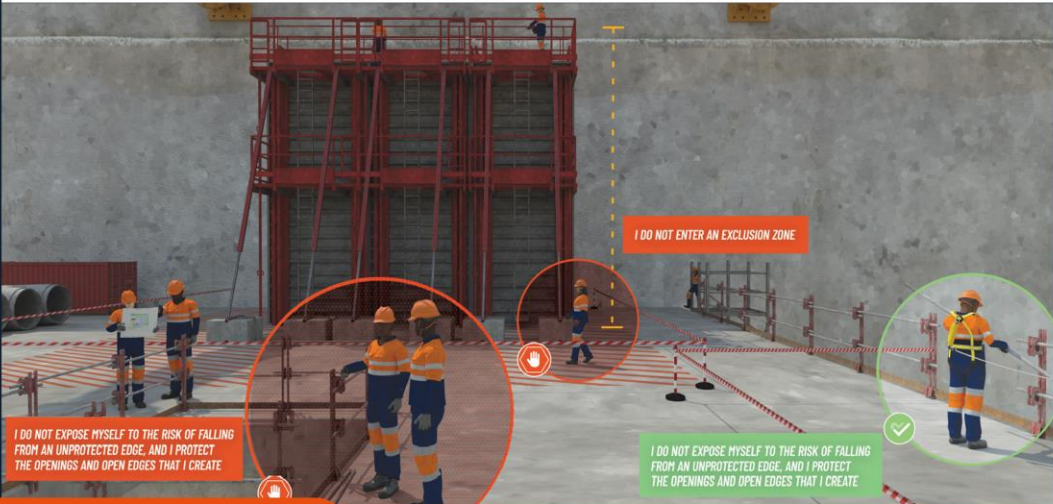
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1. Life-Saving rules



RISK WORK AT HEIGHT

- ✔ I do not expose myself to the risk of falling from an unprotected edge, and I protect the openings and open edges that I create
- ✔ I do not enter an exclusion zone



If unable to adhere to these rules, I "STOP" and alert my hierarchy.

SAFETY
CULTURE



RISK STABILITY

- ✔ I stabilise materials, plant and equipment at all times, and I ensure that the support they rely on can take the load
- ✔ I do not enter an exclusion zone



If unable to adhere to these rules, I "STOP" and alert my hierarchy.

SAFETY
CULTURE

2. Securing walkways from fall from height

2. 1. Protection of floor openings

The protection of the hoppers is carried out in accordance with the instructions of the methods department

- Floor openings must imperatively be protected and indicated
- Guards must be fixed, solid and maintained
- The hopper guards are installed in such a way that there is no difference in height greater than 21 mm. If it is technically impossible to comply with the 21 mm requirement, a visual marking of the difference in height is applied.

The passage of construction machinery (BY, subcontractors, co-contractors) over floor openings is necessarily subject to validation by the Methods Department

2.2. Securing fragile surfaces

Floors and fragile surfaces, not resistant to the passage of a person (resistance less than 150kg/m²), located even temporarily in the work area or paths, must be:

- Protected by a collective protection preventing any fall through the surface
- Identified by floor markings and a fall hazard display near the area
- Identified as prohibited for stepping and for storage of equipment



3. Securing two different heights

3. 1. Collective protection - handrails

Any difference in height between 2 levels and presenting a risk of falling is subjected to a risk analysis in order to determine the appropriate preventive measures.

In all cases, any difference in height $\geq 50\text{cm}$ between 2 levels requires the installation of appropriate collective protection.

For collective protection ensure :

- A "upper" protection against falls from a height between 100 and 110 cm
- A "lower" protection with a height of approximately 15 cm leaving no space between the protection and the work surface
- An "intermediate" protection leaving a free space of 47cm maximum either with the toe board and with the top rail
- Sufficient mechanical protection to prevent the fall of a person or a machine as needed.
- They are made of metallic material treated against corrosion (if demonstrated technically impossible, then the use of any other material must be validated by Methods and H&S departments)

The installation of collective protection shall:

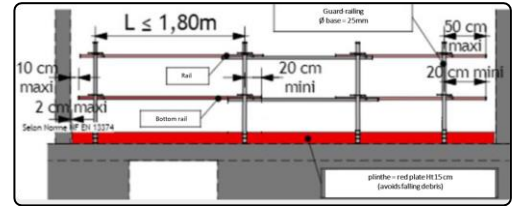
- Be performed from a safe workstation
- Be based on an agreed design which integrates as much as possible the constraints of all the stakeholders concerned, in order to avoid the temporary removal of these protections
- Follow the evolution of the height of the work surface to guarantee at all times the protection against falls in line with the requirements of collective protection

It is forbidden to pass through, step over, lean on or hang equipment on a classic collective protection (rail - under rail - plinth)

Suitable grating/mesh can also be positioned against the handrailings (in particular to limit the risk of dropped objects)

The protection of top shafts is ensured by the installation of heavy protection sized by the Methods Department (including mesh protecting from falling object).

The dimensioning takes into account the forces linked to a collision with a machine, and, if necessary, for the equipment to be fixed to it.



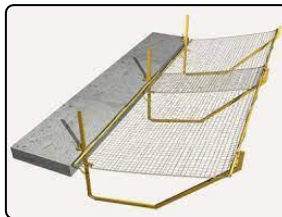
3.2. Safety nets

For works on roofs and only when the installation of collective protection such as handrailings is technically impossible, protection against falls is ensured by the use of safety nets, which are :

- Compliant with applicable regulations and standards (for example EN 1263-1 in Europe)
- Dimensioned by a competent person on the basis of a calculation note
- Installed by competent personnel in accordance with the manufacturer's instructions and the local regulations in force (design, installation/removal, verification, maintenance)
- Checked at least once a week by a competent person (good state of preservation, tension, fixing systems, etc.) **and also when weather conditions require it.**

The installation of a fall safety net can also be provided in addition to the guardrails, in particular to protect against dropped objects.

Validation by the Methods department must be carried out for the entire system (platform + net).



4. Safe access means

Vertical paths are made in order of priority with :

- Ramps (in all cases secured with collective protection when height between the 2 levels is $\geq 50\text{cm}$)
- Permanent, temporary or existing stairs (maintained in good condition)
- Mechanized means (elevators etc.)

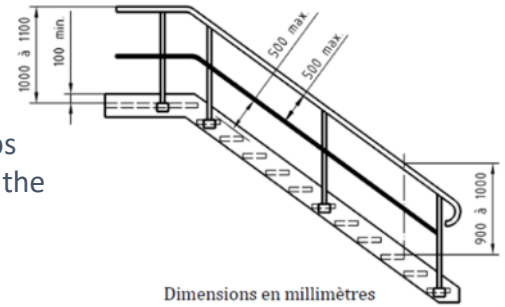
If it is impossible to install stairs or mechanized means, access may be made by fixed ladders.



4.1. Stairs and ladders with steps

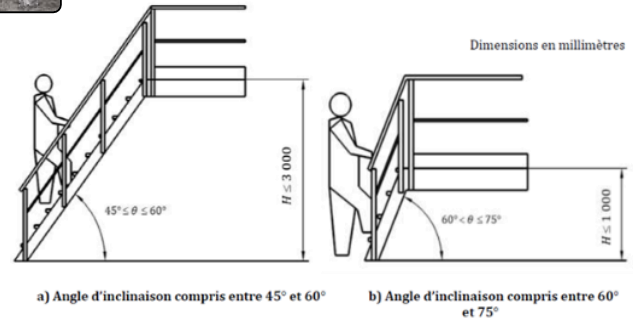
Temporary stairs are made of metal materials treated against corrosion and their installation complies with the following provisions:

- A staircase must have at least one handrail
- Two handrails If the width of the staircase is ≥ 1200 mm
- Ladders with steps must have two handrails
- Intermediate landing for any staircase with more than 25 steps
- Stability of temporary stairs to be ensured according to the supplier's instructions



Ladders with steps respect the following requirements :

- Identical steps height
- Handrails on both sides
- Maximum slope of between 45° and 75°
- Stability of step ladders to be ensured according to the supplier's instructions
- flight height depending on the stair angle



The landing area is at the level of the upper step

4.2. Access tower and modular staircase

This equipment :

- Meet collective protection requirements
- Are installed according to the manufacturer's assembly plan by competent personnel. This plan is available on site and respects:
 - The use of lock nuts
 - A lashing device at the head
 - Wooden wedging secured to the feet (screws, lag bolts, no nails)
- Are approved by competent personnel
- Are subject to a suitability review
- Are subject to daily and quarterly verifications

The checks and dates are displayed on the equipment (scafftag type).

The maximum number of people per landing area is indicated on the equipment and respected at any time.

Prioritise craneable access towers: if necessary, the lifting points are identified and appear on the assembly plan



4.3. Specific accesses

Access to precast panels

If it is impossible to use a MWEF, use a temporary ladder allowing:

- to safely hook and unhook the prefabricated element
- to implement the pouring of concrete in case of "double walls"
- This access also allows the installation of guardrails at the head of the walls before the construction of the slabs
- Fixing hooks (not "homemade") must be adapted to the width of the wall – refer to BYCN Material



Access on slab

If it is impossible to set up a metallic staircase, use a TEMPORARY telescopic access mean on the slab with automatic closure access gate

Free ladders are prohibited



Adjustable plate allowing to rest and to hook equipment to the slab

Access to excavations



"Worksite" footbridges

They are dimensioned by the Methods dpt considering:

- Their use (pedestrian bridge, passage of heavy vehicles, etc.)
- Operating load per m^2 + max load

- Footbridges equipped with handrails + toe boards
- No storage on footbridges
- Slope 15% maximum and non-slippery (pay attention to the specific requirements for temporary "public" footbridges according to the standards for disable persons/wheelchairs)



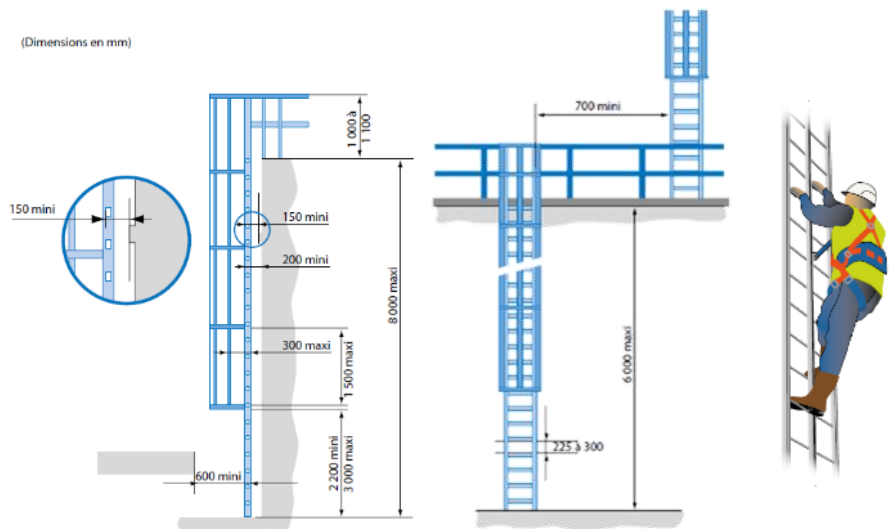
4.4. Fixed ladders

The ISO 14122-4 standard (unless the applicable local regulations or standards are more demanding) serves as a reference for the safety measures applicable to fixed ladders allowing access to any work platform.

Refer to the manufacturers and the Methods department for specific cases.

Any access by fixed ladder is authorized if it:

- is an integral part of the equipment (scaffolding, crane, etc.) or is totally attached to the interior of the structure or equipment with in all cases a security at each level
- is fitted with a safety cage (Ø between 650 and 800 mm) for any vertical access ladder greater than 3 m. The 1st arch is between 2.20m and 3m from the ground
- is provided with top hooks allowing it to be secured on a hanging bar
- is fitted with antislip pads and grap rails for access to terrace in existing buildings (skydome, hatch)
- Is fitted with a permanent fall arrest system for any fixed vertical ladder longer than 8m or, by default, access is done by using a personal fall arrest system (in both cases, wearing harness is mandatory)
- As soon as the difference in level is greater than 8 m, the installation of landings with a change of flight every 6 meters is required



Only one person at a time is authorized on a fixed ladder

Ladders can have a front or side exit to the finish area. The width of the outlet opening must be between 500 mm and 700 mm

Specificities for exit/access gate:

- The opening of the gate must not be towards the outside of the platform
- Design in such a way as to allow easy opening
- Automatic closing (springs or gravity)
- Fitted with handrail and middle rail

Specificities for exit / access hatch:

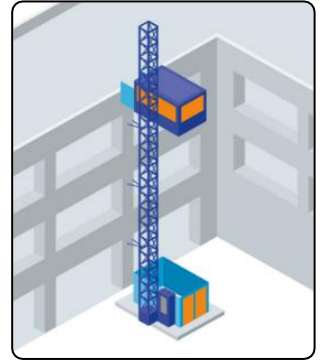
- Opening at least equal to the dimensions required for the ladder's safety cage
- The hatch must only move upwards or horizontally
- Manual and easy opening
- Allows the safe passage of the operator and an automatic closing without the intervention of the operator (springs, hydraulic means)

4.5. Construction elevator

The EN 12159 standard (unless the applicable local regulations or standards are more demanding) serves as a reference for the safety measures applicable to construction site elevators intended for use by personnel, serving landings and having a shaft. Refer to BYCN Material about minimum height requiring the installation of elevator, including for tower crane elevators.

The following requirements apply:

- The cage must have a roof
- Anchoring in accordance with the manufacturer's recommendations and validated by Methods
- Presence of instructions and safety plates (display of max load, maximum number of people, etc.)
- interlocked access to prevent crushing when descending
- Regulatory landing door at each level, opening only in the presence of the cage
- Landing doors with self-locking device (without means of bypass)
- Under normal operating conditions, it should not be possible to start the cage or keep it moving unless the landing doors are all closed and locked.
- Users must be informed about the instructions for use, safety instructions and emergency instructions. They can use the elevators themselves with floor call button and automatic door unlocking
- The elevator installer must carry out or have carried out static and dynamic tests to ensure that the construction elevator has been assembled correctly and to check that all the devices provided are present and operate correctly.
- Pre-use inspections of this equipment should be carried out on site by an Inspection Body certified against ISO 17020 and accredited by an Accreditation Organism member of (either):
 - The International Laboratory Accreditation Cooperation (ILAC)
 - The International Accreditation Forum (IAF)
 - The European cooperation for Accreditation (EA)
 - Alternatively, by another equivalent control office
- Elevators without call buttons or automatic door unlocking require an elevator operator, who must be trained by the supplier and authorized by the employer
- Emergency management:
 - The cage must have at least one door or hatch through which an emergency exit is possible. This door/latch shall be opened without any key from outside the cage and with a special key from inside the cage
 - Possibility to unlock the landing doors from the landing side using a special key
 - The locking of the emergency exit door must be controlled by an electrical safety device. This device must stop the lift when the locking is no longer effective. Re-commissioning of the elevator shall only be possible by an intentional re-locking of the emergency exit



Securing elevators bay

A protective grid (GEMAGRILLE® type) is installed to close off the elevator shafts, allowing full-height protection and allowing the lift company to intervene safely.

- Studs in the lower part inserted into the concrete (pre-drilling of the holes)
- Safety counter nut blocking the tightening nuts
- Positioning of the upper element against the opening top
- Tightening of the threaded rods to stiffen the grille to the correct opening dimension



A classical handrail protection is not authorized

4.6. Podiums for staircase and elevator

A validation by the Methods Department is mandatory:

- Distribution of loads
- Sizing of shoes, fixing dowels and implantation...
- Removal of anchors

Storage on podiums is prohibited



4.7. Goods lifts

Goods lifts/material lifts are reserved exclusively for the supply of materials and equipment.

It is mandatory to provide:

- Types of anchors sized by the manufacturer and checked by the Methods department
- Mandatory instructions and safety plates
- Maximum load
- Regulatory landing door at each service level that cannot be opened without the presence of the cabin
- Peripheral protection at the base
- Mandatory training by the lift installer
- Authorization by the employer
- Pre-use inspections of this equipment should be carried out on site by an Inspection Body certified against ISO 17020 and accredited by an Accreditation Organism member of (either):
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5. Temporary removal of protection means

The first approach is to avoid the withdrawal of collective protection. There are specific devices for transporting equipment onto protected floors while guaranteeing permanent collective protection for the personnel.

Tilting barriers can be installed on any type of floor (including scaffolding) as long as the load bearing is validated by the Methods Department.

In the absence of these devices, the temporary withdrawal of collective protection implies the immediate implementation of compensatory measures (marking, surveillance, etc.) which must be anticipated during the preparation of activities through method statements and risk analyses.

Any intervention is then carried out by the use of individual protections.

At the end of the intervention, the collective protection is returned to its initial state of conformity.



6. Temporary work at height equipment

Work at height is carried out from stable, fixed or mobile working platforms or from scaffolding equipped with collective protection (top rails, mid-rails and toe boards).

These equipments are made of metallic parts, except for use near electrical installations.

They are secured by rigid guardrails on their 4 sides and comply with the rules of collective protection.

Scaffolding and all platforms used for work at height are equipped with a net or physical protection to prevent falling objects. In default, an exclusion zone is implemented in accordance with the height and the risk of deflection of the object. The portable tools are then also hung via straps.

All use of such equipment requires :

- identification of the need and a risk analysis
- Verification of the suitability of the equipment used for the access at height/task to be performed
- Installation on a flat and suitable surface
- Deployment and locking of stabilization devices if required

6.1. Mobile Working Elevating Platform (MEWP)

Only specially trained and authorized personnel is allowed to use this type of equipment.

- Wearing a compliant harness is mandatory, except in the case of a scissor lift (type 2A, 3A)
- The harness must be connected to the dedicated anchor point specified in the manufacturer’s guide. The lanyard does not exceed 1,5m (if not, use an retractable lineyard)
- Marked exclusion zone on the ground including the area under the basket
- Presence on the ground of a person having a driving authorization adapted to the type of platform used. This person acts as a watchman and is trained to be able to secure the MEWP from the ground and to rescue the users (the MEWP must be positioned in such a

	Type 1 (Travels only in stowed position)	Type 2 (Chassis controls only, can drive elevated)	Type 3 (Both Chassis and platform controls, can drive elevated)
Group A (Scissor Lift Types) Platform does not extend beyond tipping axis	Type 1A 	Type 2A, 3A 	
Group B (Boom Lift Types) Platform can extend beyond tipping axis	Type 1B 	Type 2B, 3B 	
		UBIT: Under Bridge Inspection Trucks Type 2-3 B 	
VM (Vehicle mounted bucket truck)	VM 		

- way as to guarantee free access to its secondary control pannel located at ground level
- MEWP is stopped in the event of wind exceeding the limits provided by the manufacturer

Note: for the moving of telehandlers, refer to the document **Signaling standards and vehicle-pedestrian protection systems (BYTP-H&S-INF-2168)**

6.2. Fixed and mobile scaffolding, shoring towers

The design, assembly, modification, inspection and dismantling of fixed and mobile scaffolding are carried out by competent persons.

This equipment is equipped with:

- Handrails
- Joint and secured floor elements
- A secured access system (internal fixed ladder)
- Self-closing hatches not overlapping from one level to another

Compulsory installation of a handrail at 1.50 m to the right of the ladder



Hatches are not:

- be superimposed
- To stay open

Preference should be given to scaffolding which, by its design, guarantees assembly and disassembly with collective protection in place. Failing this, their assembly, modification and disassembly are carried out with the use of individual protections, via identified anchoring points that comply with the requirements.

The assembly-disassembly is always carried out according to the assembly/disassembly kinematics of the manufacturer. This plan is available on site and respects:

- The use of lock nuts
- A lashing device at the head
- Wooden wedging secured to the feet (screws, lag bolts, no nails)

Any scaffolding higher than 24m requires a strength and stability calculation note validated by the Methods Department. Ditto in the case of nets, reception of materials and/or winches affixed to the scaffolding.

The admissible load of a scaffolding must be visibly indicated on the scaffolding.

The compliance of the scaffolding or the prohibition to access it is visibly indicated at the access points of the scaffolding, including a temporary prohibition (red / green scaffolding system or equivalent).

A complete inspection of the scaffolding must be carried out imperatively after any period of severe weather (strong winds, storm) before its return to service.

In case of damage or alteration, the highest priority is to immediately quarantine the scaffolding and prohibit access until it is brought back into compliance.

Regarding the frequency of control:

- **Inspection after repair: A complete inspection control must be systematically carried out after any significant repair or modification to ensure the resumption of compliance and the safety of the work.**
- **Inspection before repair: An inspection of the initial state of damage may be required as part of an investigation (HiPo type) or for insurance purposes (documentation and traceability).**

6.3. Platforms on mast(s)

The use of mast platforms is managed in coordination with the Materiel department

Platforms on mast(s) are temporary installations intended to work at different levels on building facades.

No access can be made to a building with a platform on a mast

Installation on site:

- Bearing capacity of the ground to be checked
- Regulatory verification at commissioning stage (or in the event of modification) then every 6 months should be carried out on site by an Inspection Body certified against ISO 17020 and accredited by an Accreditation Organism member of (either):
 - The International Laboratory Accreditation Cooperation (ILAC)
 - The International Accreditation Forum (IAF)
 - The European cooperation for Accreditation (EA)
 - Alternatively, by another equivalent control office
- Any structural modification must be carried out and validated by the manufacturer and the installer

Staff training :

- User's training is based on the manufacturer's operating instructions
- Authorization of use issued by the project manager

Use of the platform:

- Platform not be used for access or delivery of material
- Load capacity/max number of people is clearly displayed
- The organization of rescue (platform blocked at height) is anticipated and the measures planned



Training and authorization to use are mandatory

6.4. Cantilever work platforms

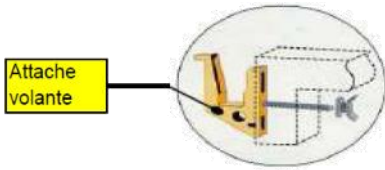
Arrange on site to have:

- Layout plans
- Manufacturer's assembly and operating instructions
- Verification checklists before each shift

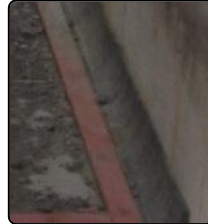
Consoles:

Unhooked from the crane when the consoles are:

- Placed on the fasteners (quantities and positions defined by the Methods Department)
- Equipped with anti-lifting and anti-rolling safety system
- Console > 2m wide = 3 fasteners
- Compulsory sealing flaps

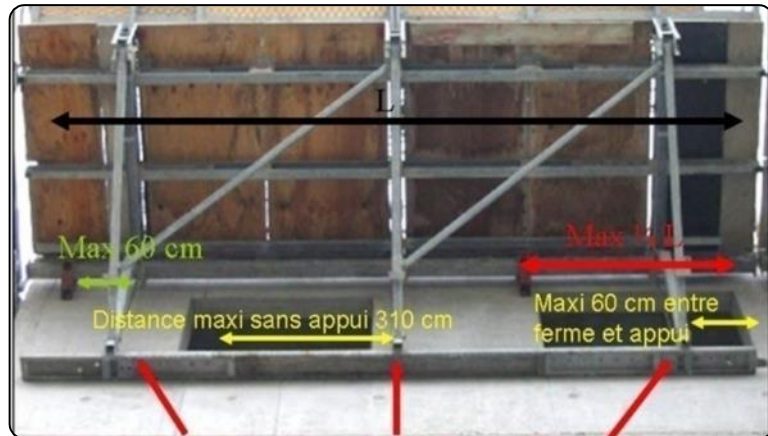


Sealing flap



Prohibition to lift a platform with a deformed lifting ring

Mandatory verification/approval after initial assembly and then quarterly by a competent person of the material and by Temporary equipment responsible person

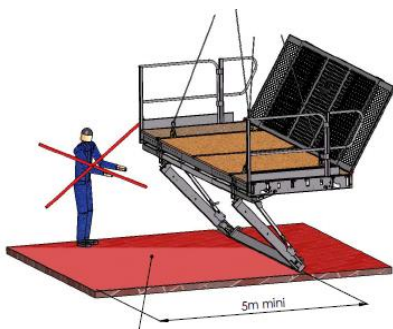


One hitch per truss

CMU display. Do not exceed the maximum storage capacity of the console (no bulky storage: water, oil, steel, etc.).

Any use in material receipt is validated by the Methods Department

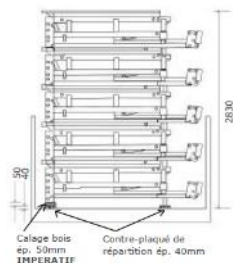
Folding



Storage / Loading-unloading

Favor stacking on axis

Refer to the manufacturer's instructions



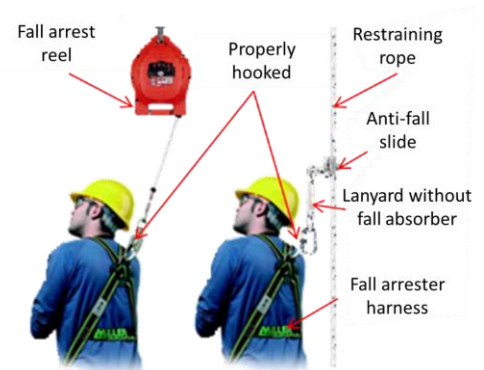
6.5. Suspended platforms

Refer to BYCN Materiel department or equivalent material department

A variable-level suspended platform is a temporary work equipment consisting of a platform moving using winches and along cables which are connected to a suspension device.

The platforms:

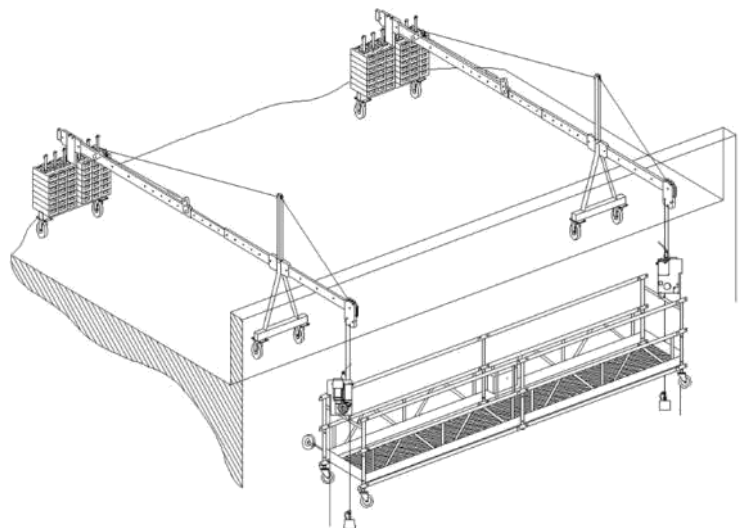
- Are equipped with an overload device
- Are equipped with devices against shocks in case of contact with the facade
- Cannot be used in adverse weather conditions (high winds, etc.). See the manufacturer's instructions depending on the type of equipment installed
- Mandatory verification after assembly (platform, jib, counterweight, etc.)
- Load capacity is clearly displayed
- Individual fall arrestor must be installed at the head of the building
- The organization of rescue (platform blocked at height) is anticipated and the measures planned



The platform cannot be used by a single user

Winches:

- Winches are certified and regularly checked
- Each winch uses 2 cables: 1 for work and 1 for safety
- The winches are equipped with safety devices (parachute system)
- Winches can be manual, electric, pneumatic or hydraulic



Training and authorization to use are mandatory

6.6. Individual mobile platform (light)

- Use by personnel aware about their use and the checks to be carried out before use
- The guardrails are rigid on the 4 sides (chain prohibited) and correctly engaged
- Presence of non-slip pads
- it is forbidden to work from the steps of the platform
- Go up and down facing the platform
- Preferably the platforms are used in the direction of the work



The use of free stepladders is prohibited

6.7. Temporary decking and walkways

- The sizing of decking is carried out by the Methods department and approved by Temporary equipment responsible person
- Assemble on the ground the decking/walkways
- Collective protection must be put in place before final installation
- The maximum slope allowed is 15%
- Vigilance regarding the laying of the decking with the risk of falling from height
- A non-slip coating is put in place (carpet, expanded metal, etc.)
- A regular check of the condition of the wood must also be carried out



6.8. Works on ropes

The use of rope access work is a last resort, when no other technical solution allows the work to be carried out (to be demonstrated and documented).

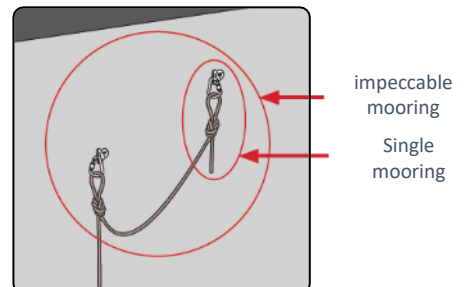
- Users shall comply with applicable local and international regulations
- An agreement from the local authorities may be necessary before undertaking any work on ropes
- Operations shall be subcontracted only to accredited companies ensuring appropriate skilled/competent personnel
- Specific procedures and risk analyzes must be carried out for each operation

Particular attention will be paid to the following elements:

- The equipment is in compliance with the applicable standards (local and international such as IRATA code - *International code of practice for industrial rope access*)
- Stakeholders with recognized certified training, such as IRATA, SPRAT, CQP. Or training according to the applicable standards
- Equipment and tools must be permanently attached
- Exclusion zone is marked on the ground
- Display/waring signage is made that such activity is ongoing



A contingency plan specific to these operations must be drawn up. The installation of the work ropes can in particular integrate disengageable systems allowing evacuation from the ground. These devices must be tested before the start of work.



7. Personal Protective Equipment (PPE)

The use of personal protection systems against falls from a height is the last choice (apart for use in a MEWP) . In case of use, wearing a full body harness connected to a certified anchor point is compulsory.

Any method statement requiring the use of a safety harness (excluding for use in a MWEF) must be validated by the project H&S department

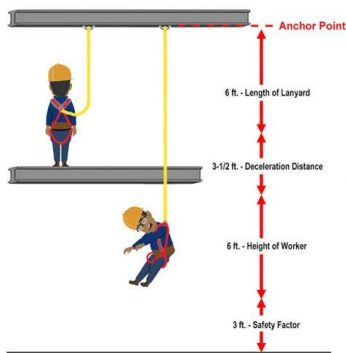
Personal fall protection systems:

- Compliant with regulations, applicable standards (ANSI, EN, AS, CSA, etc.)
- Used under the conditions and limits recommended by the manufacturer
- Only the integral harness is accepted and the carabiners must have automatic locking and double action opening
- The harnesses are equipped with 2 lanyards in order to be able to move around in complete safety, without ever having to be fully disconnect
- It is strongly recommended to use harnesses fitted with "footrest" systems which - deployed after a fall - help to restore blood circulation, particularly in the area of the leg straps, by relieving the user of his own weight
- The use of individual protection (harness, lanyard) for work at heights is strictly reserved for staff who have undergone specific training, and a list of these authorized personnel must be established and regularly updated by the employer.
- Each harness is assigned to a single person (harness not shared). In the event that individual allocation is impossible, a control process must be put in place before each use of the harness



Anchor points and lifelines:

Prioritize the installation of a fall restraint system, ie a system preventing the worker from reaching any point where a risk of falling has been identified.



The fall arrest system (anchor point/lifeline, lanyard, absorber if necessary, harness) must take into account the clearance required in the event of a fall. The calculation of the clearance takes into account the positioning of the anchor point (ideally located above the head of the worker), the length of the lanyard, that of the deployed absorber, the size of the the person, as well as an additional safety distance of 1m.

Take into account the pendular movement of the suspended body following a fall and due to the distance between the anchor point and the point of fall.

Anchor points and lifelines (horizontal and vertical) must comply with applicable regulations and standards, installed and checked by a competent person according to the prescriptions of the manufacturer and according to the design defined by the Methods Department (standard or specific). Anchor points, including temporary ones, are certified by trained and competent personnel.

Warning: The use of cable clips (bulldog) in the design of a lifeline is prohibited



8. Bolted assemblies

Bolted assemblies subjected to fatigue constraints, whether assembled on-site or pre-assembled, must be known and identified.

They must be:

- Confirmed compliant with assembly instructions (type of bolts, number, tightening torque, etc.)
- Regularly inspected to identify any risks related to material fatigue (corrosion, cracks, etc.)
- Replaced as necessary

The assembly, commissioning, and maintenance procedures defining the criteria for material acceptable conditions must be implemented according to projects' organisations.



9. Forbidden equipment for work/access at height

- Basket/cage lifted by the forks of a forklift
- Basket/cage lifted by the arm of a telehandler (even if there is a coupling device instead of forks)

